

Annex 176C

(normative)

Chip-to-chip Attachment Unit Interfaces 200GAUI-1 C2C, 400GAUI-2 C2C, 800GAUI-4 C2C, and 1.6TAUI-8 C2C

176C.1 Overview

This annex defines the functional and electrical characteristics for the optional chip-to-chip 200 Gb/s one-lane Attachment Unit Interface (200GAUI-1 C2C), 400 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2C), 800 Gb/s four-lane Attachment Unit Interface (800GAUI-4 C2C), and 1.6 Tb/s eight-lane Attachment Unit Interface (1.6TAUI-8 C2C). These interfaces have specified electrical characteristics, and may optionally be used when designing systems with electrical interconnects.

200GAUI-1 C2C, 400GAUI-2 C2C, 800GAUI-4 C2C and 1.6TAUI-8 C2C are physical instantiations of the PMA service interface, as described in 176.2, which enable partitioning a Physical Layer implementation across multiple devices. Within this annex, these interfaces are collectively referenced as 200 Gb/s per lane AUI-C2C.

Figure 176C–1 shows the relationship of the 200 Gb/s per lane AUI-C2C, along with other Ethernet sublayers, to the ISO/IEC Open System Interconnection (OSI) reference model. Physical Layer partitioning options using 200 Gb/s per lane AUI-C2C are provided in Annex 176B.

This annex defines specifications for two classes of C2C transmitters and two classes of C2C receivers, class A or class B, to allow for a range of package characteristics. A transmitter or receiver is classified as class A or class B if it meets all of the specifications for class A or class B, respectively. The required characteristics of the electrical interconnect between two devices depend on the transmitter class on one device and the receiver class on the other device.

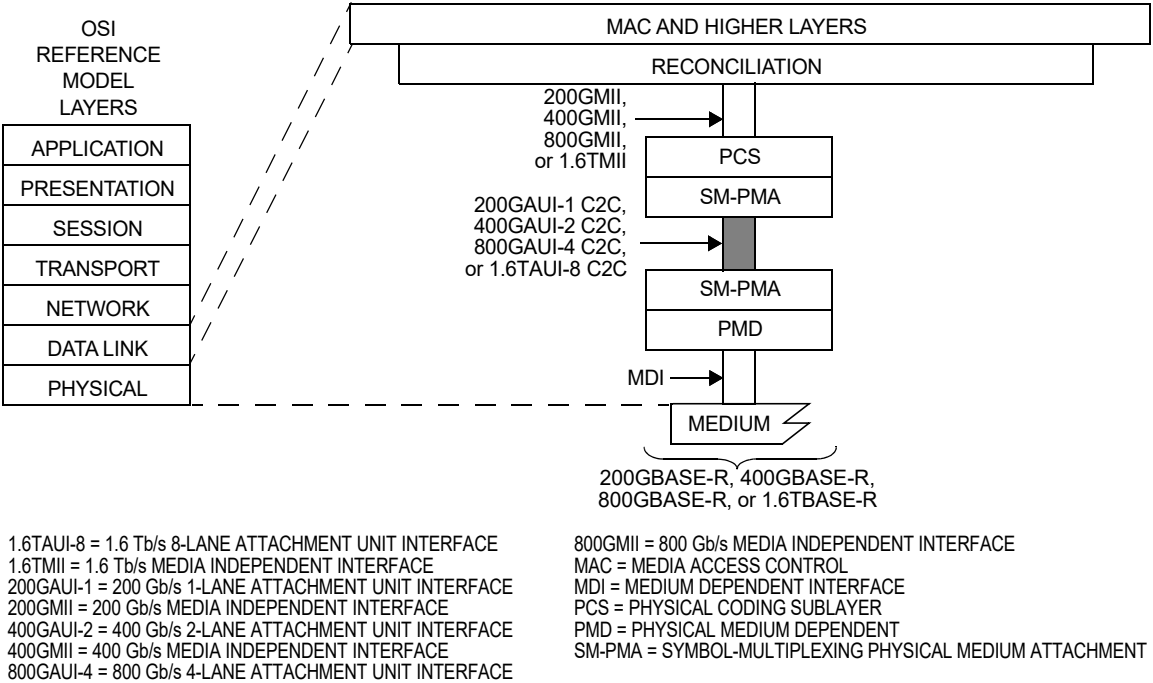


Figure 176C-1—200 Gb/s per lane AUI-C2C relationship to the ISO/IEC Open System Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model

176C.2 Error Ratio Allocation

With a compliant input signal, a C2C component receiver (see 176C.6.4) is expected to meet the block error ratio (BLER) of 1.45×10^{-11} (see 174A.6), measured at the PMA adjacent to the C2C component using the method described in 174A.9, with BER_{added} equal to 2.841×10^{-4} .

176C.3 Functional specification

A 200 Gb/s per lane AUI-C2C is a bidirectional data path. Each direction of a 200GAUI-1 C2C, 400GAUI-2 C2C, 800GAUI-4 C2C or 1.6TAUI-8 C2C data path contains one, two, four, or eight lanes, respectively

The 200 Gb/s per lane AUI-C2C bidirectional link is described in terms of two C2C components that are connected by a C2C channel.

An n -lane C2C component is functionally equivalent to a corresponding n -lane PMD specified in Clause 178. The C2C component shall meet the functional specifications in 178.8 unless stated otherwise.

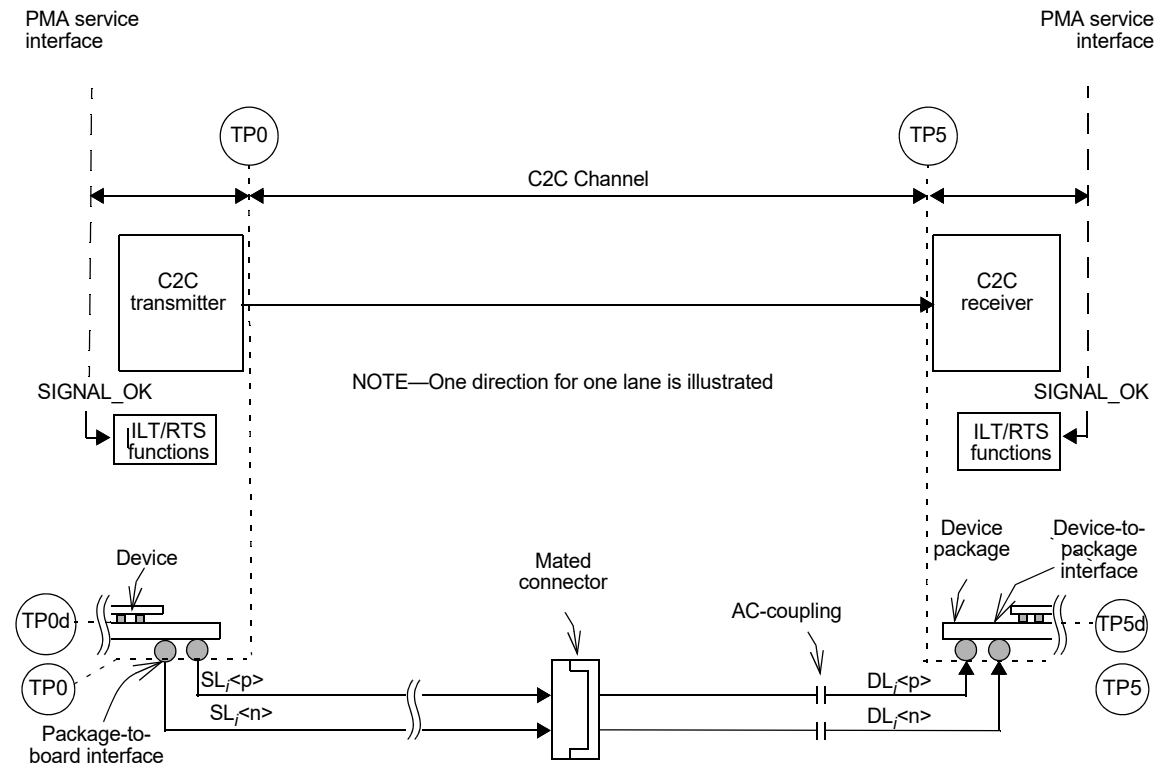
NOTE 1—As part of the functional equivalence to a PMD, C2C components include the inter-sublayer link training (ILT) function with E1 format and the ready-to-send (RTS) function, specified in Annex 178B.

The C2C channel is specified in 176C.7.

The electrical characteristics of the C2C components are specified in 176C.6.3 and 176C.6.4.

NOTE 2—The electrical specifications of C2C components are not identical to those of the corresponding PMDs defined in Clause 178.

Figure 176C–2 depicts a typical 200 Gb/s per lane AUI-C2C application.



NOTE 1—The number of lanes n is equal to 1 for 200GAUI-1, 2 for 400GAUI-2, 4 for 800 GAUI-4, and 8 for 1.6TAUI-8.
NOTE 2—The source lane (SL) signals $SL_i<p>$ and $SL_i<n>$ are the positive and negative sides of the transmitter's differential signal pair on lane i and the destination lane (DL) signals $DL_i<p>$ and $DL_i<n>$ are the positive and negative sides of the receiver's differential signal pair on lane i .

Figure 176C–2—200 Gb/s per lane AUI-C2C link diagram

176C.4 Service interfaces

The SM-PMA above the 200 Gb/s per lane AUI-C2C is any $m:1$ PMA for 200GAUI-1, $m:2$ PMA for 400GAUI-2, $m:4$ PMA for 800GAUI-4, and $m:8$ PMA for 1.6TAUI-8, as specified in Clause 176. The PMA below the 200 Gb/s per lane AUI-C2C is any $1:n$ SM-PMA for 200GAUI-1, $2:n$ SM-PMA for 400GAUI-2, $4:n$ SM-PMA for 800GAUI-4, and $8:n$ SM-PMA for 1.6TAUI-8, as specified in Clause 176.

The service interface above and below the 200 Gb/s per lane AUI-C2C is the PMA service interface as specified in 176.2.

For an AUI component above the AUI channel, the SIGNAL_OK parameter of the PMA:IS_SIGNAL.indication primitive corresponds to the variable rts_status of the RTS function, as defined in 178B.8.2.1. When SIGNAL_OK is either IN_PROGRESS or FAIL, the rx_symbol parameters of the PMA:IS_DATAUNIT_i.indication primitive on all lanes are unspecified.

For an AUI component below the AUI channel, the SIGNAL_OK parameter of the PMA:IS_SIGNAL.request primitive corresponds to the variable rts_status of the RTS function, as defined in 178B.8.2.1. When SIGNAL_OK is either IN_PROGRESS or FAIL, the tx_symbol parameters of the PMA:IS_DATAUNIT_i.request primitive on all lanes are unspecified.

The service interfaces are illustrated in Figure 176C–3.

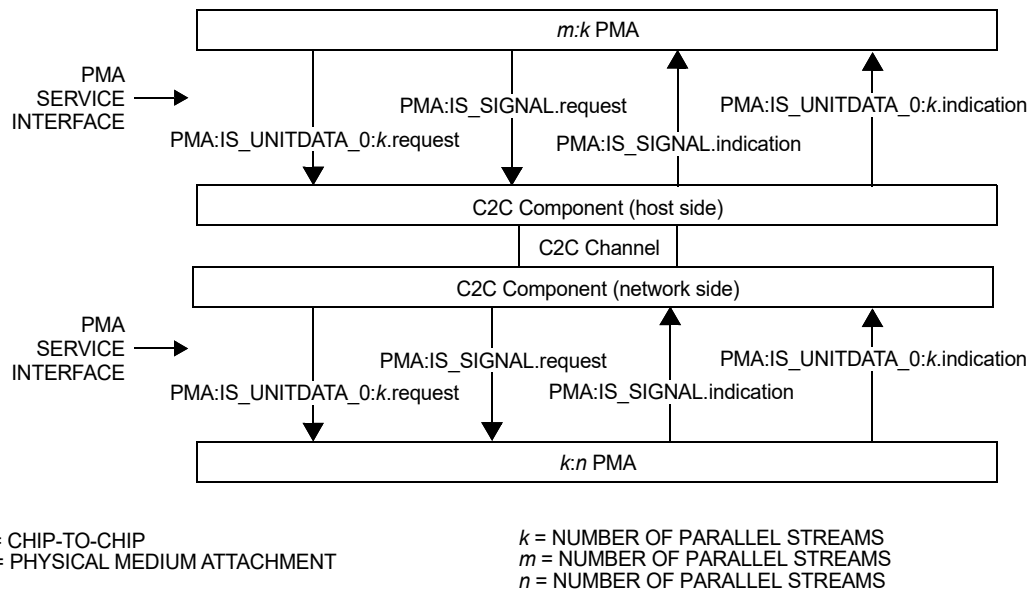


Figure 176C–3—Inter-sublayer service interfaces for a 200GAUI-1, 400GAUI-2, 800GAUI-4, or 1.6TAUI-8 C2C

176C.5 Delay constraints

The sum of the transmit and receive delays at one end of the link contributed by a 200 Gb/s per lane AUI-C2M component shall be no more than the maximum delays listed in Table 176C–1.

Descriptions of overall system delay constraints can be found in 116.4 for 200GAUI-1 and 400GAUI-2, in 169.4 for 800GAUI-4, and in 174.4 for 1.6TAUI-8.

Table 176C–1—Delay constraints

Component	Maximum (bit times) ^a	Maximum (pause_quanta) ^b	Maximum (ns)
200GAUI-1	9 216	18	46.08
400GAUI-2	18 432	36	46.08
800GAUI-4	36 864	72	46.08
1.6TAUI-8	73 728	144	46.08

^a One bit time is equal to 5 ps for 200GAUI-1, 2.5 ps for 400GAUI-2, 1.25 ps for 800GAUI-4, and 625 fs for 1.6TAUI-8. See 1.4.160 for the definition of bit time.

^b One pause_quantum is equal to 2.56 ns for 200GAUI-1, 1.28 ns for 400GAUI-2, 640 ps for 800GAUI-4, and 320 ps for 1.6TAUI-8. See 1.4.459 for the definition of pause_quanta.

176C.6 Electrical characteristics

176C.6.1 Compliance points

The electrical characteristics for C2C components are defined at test points TP0v and TP5v, described in 178.9.2.1 and 178.9.3.1.

Some of the measurements at the compliance points are compared to calculated reference results that account for a reference package, using the procedure in Annex 163A but replacing the COM-related calculation in Annex 93A with those of Annex 178A. The reference package parameters for TP0v and TP5v measurements are selected from Table 176C–9 according to the transmitter class and receiver class, respectively, to which a device claims compliance.

176C.6.2 Reference impedance

The reference impedance for differential specifications is 92.5 Ω . The reference impedance for common-mode specifications is 23.125 Ω .

NOTE—Calculation of modal S-parameters may require transformation of the measured single-ended S-parameters to the specified reference impedance. See 178A.1.3.

176C.6.3 Transmitter characteristics

The transmitter on each lane shall meet the specifications at TP0v (see 178.9.2.1) given in Table 176C–2 and detailed in the referenced subclauses.

Unless otherwise specified, transmitter or pattern generator signal measurements are made for each lane separately using a fourth-order Bessel-Thomson low-pass response with a 3 dB bandwidth of 60 GHz, with AC-coupled connection from TP0v to 50 Ω single-ended loads in the test equipment.

Measurement of the DC common-mode voltage is made with a high-impedance connection to TP0v where each conductor of the differential pair is AC-coupled to a 50 Ω termination.

A C2C component's transmitter on each lane shall meet the specifications given in Table 176C–2.

Table 176C–2—Transmitter electrical characteristics at TP0v

Parameter	Reference	Value	Units
Signaling rate (range)	176C.6.3.1	106.25 ± 50 ppm	GBd
Differential peak-to-peak output voltage (max)	176D.8.2	0.035	V
Transmitter disabled		1	V
Transmitter enabled			
DC common-mode voltage (range)	176D.8.2	0.2 to 1	V
Low-frequency peak-to-peak AC common-mode voltage, $V_{CM_{LF}}$ (max)	176C.6.3.3	0.032	V
Signal to AC common-mode noise ratio, $SCMR$ (min)	176C.6.3.4	15	dB
Difference effective return loss, $dERL$ (min)	176C.6.3.7	–3	dB
Common-mode to common-mode return loss, RL_{cc} (min)	178.9.2.4	3.25	dB
Difference steady-state voltage, dv_f (range)	176C.6.3.6	0 to 0.1	V
Difference linear fit pulse peak ratio, dR_{peak} (min)	176C.6.3.6	0	—
Level separation mismatch ratio, R_{LM} (min)	179.9.4.3	0.95	—
Transmitter output equalization			
absolute value of step size range for all taps	179.9.4.2.4	0.005 to 0.025	—
value at min state for c(–3) (max)	179.9.4.2.5	–0.05	—
value at max state for c(–3) (min)	179.9.4.2.5	0	—
value at min state for c(–2) (max)	179.9.4.2.5	0	—
value at max state for c(–2) (min)	179.9.4.2.5	0.1	—
value at min state for c(–1) (max)	179.9.4.2.5	–0.3	—
value at max state for c(–1) (min)	179.9.4.2.5	0	—
value at min state for c(0) (max)	179.9.4.2.5	0.5	—
value at min state for c(1) (max)	179.9.4.2.5	–0.1	—
value at max state for c(1) (min)	179.9.4.2.5	0	—
Signal-to-noise-and-distortion ratio, SNDR (min)	179.9.4.6	Table 179–9	dB
Signal-to-residual-intersymbol-interference ratio, SNR_{ISI} (min)	176C.6.3.5	28	dB
Output jitter (max)	176C.6.3.8	0.023	UI
JH_{RMS}		0.025	UI
EOJ_{03}		0.118	UI
JH_{4u}			
Common-mode to differential-mode return loss, RL_{dc}	176C.6.3.9	Equation (176C–1)	dB

176C.6.3.1 Transmitter signaling rate

The transmitter signaling rate shall be within the range specified in Table 176C–2.

NOTE—For a transmitter of an AUI-C2C component in the receive direction, the signaling rate is derived from the PMD receiver input.

176C.6.3.2 Transmitter equalization settings

The transmit equalizer is identical to that specified in 179.9.4.2 and utilizes the inter-sublayer link training (ILT) function with a training frame of E1 format as defined in 179.8.9 with exception that output step size and coefficient limits are specified in Table 176C–2.

176C.6.3.3 Peak-to-peak AC common-mode voltage

The low-frequency and full-band peak-to-peak AC common-mode voltage, VCM_{LF} and VCM_{FB} , respectively, are defined by the method specified in 176D.8.2.

The low-frequency peak-to-peak AC common-mode voltages shall meet the specification for VCM_{LF} (max) in Table 176C–2.

176C.6.3.4 Signal to AC common-mode noise ratio

Signal to AC common-mode noise ratio, $SCMR$, is defined by the method specified in 178.9.2.6 with the exception that the full-band peak-to-peak AC common-mode voltage is defined in 176C.6.3.3.

The signal to AC common-mode noise ratio shall meet the specification for $SCMR$ (min) in Table 176C–2.

176C.6.3.5 Signal-to-residual-intersymbol-interference ratio

Signal-to-residual-intersymbol-interference ratio SNR_{ISI} is defined by the method specified in 179.9.4.4.

Signal-to-residual-intersymbol-interference ratio shall meet the specification for SNR_{ISI} (min) in Table 176C–2.

176C.6.3.6 Difference steady-state voltage and difference linear fit pulse peak ratio

The difference steady-state voltage dv_f , and difference linear fit pulse peak ratio dR_{peak} are defined in 178.9.2.5.

176C.6.3.7 Transmitter difference effective return loss (ERL)

The transmitter difference ERL at TP0v is computed using the procedure in 163A.3.2.2 with the values in Table 176C–3 and Table 176C–9, with differential reference impedance equal to $92.5\ \Omega$, and with the value of T_{fx} equal to twice the delay from TP0 to TP0v. The reference value, $ERL^{(ref)}$, is calculated using the device package model of the transmitter class to which the device adheres.

Table 176C–3—Transmitter and receiver ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.005	ns
Incremental available signal loss factor	β_x	0	GHz

Table 176C–3—Transmitter and receiver ERL parameter values (*continued*)

Parameter	Symbol	Value	Units
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	400	UI
Equalizer length associated with reflection signal	N_{bx}	16	UI
Number of samples per unit interval	M	32	
Tukey window flag	tw	1	—

The difference ERL at TP0v shall be greater than or equal to $dERL$ (min) specified in Table 176C–2.

176C.6.3.8 Output jitter

The output jitter parameters JH_{RMS} , JH_{4u} , and EOJ_{03} are defined in 179.9.4.7.

176C.6.3.9 Transmitter common-mode to differential-mode return loss

The common-mode to differential-mode return loss of the transmitter shall meet Equation (176C–1). The common-mode to differential-mode return loss limit is illustrated by Figure 176C–4.

$$RL_{dc}(f) \geq \begin{cases} 25 - 20 \frac{f}{106.25} & 0.05 \leq f \leq 53.125 \\ 15 & 53.125 \leq f \leq 67 \end{cases} \quad (176C-1)$$

where

RL_{dc} is the common-mode to differential-mode return loss in dB
 f is the frequency in GHz

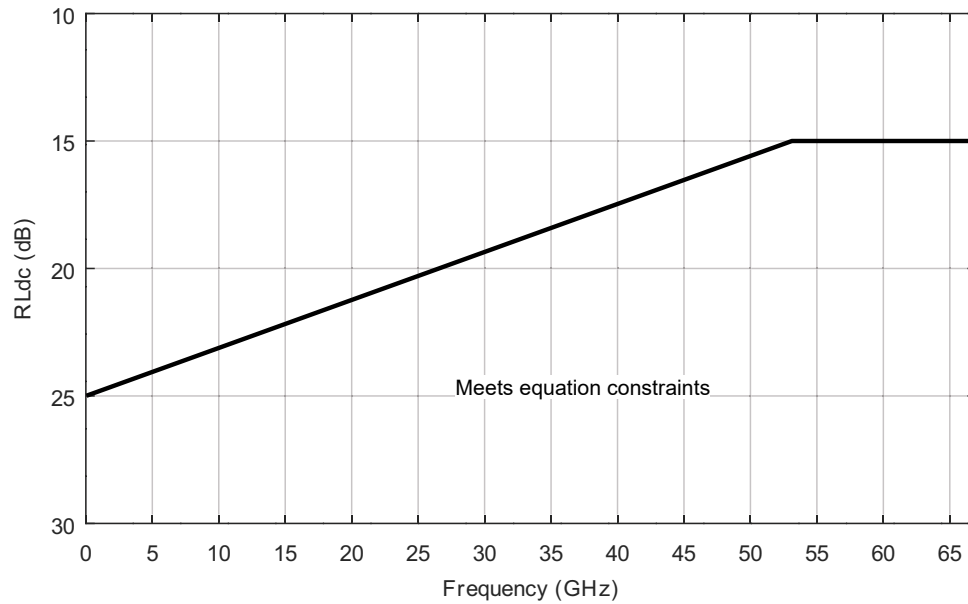


Figure 176C-4—Transmitter common-mode to differential-mode return loss limit

176C.6.4 Receiver characteristics

A receiver on each lane shall meet the specifications given in Table 176C-4

Table 176C-4—Receiver characteristics at TP5v

Parameter	Reference	Value	Units
Signaling rate (range)	176C.6.4.1	106.25 ± 50 ppm	GBd
Transmitter output amplitude tolerance	176C.6.4.7	0.5	V
Difference effective return loss, $dERL$ (min)	176C.6.4.2	-3	dB
Differential-mode to common-mode return loss, RL_{cd}	176C.6.4.3	Equation (176C-2)	dB
Interference tolerance	176C.6.4.5	Table 176C-7	—
Jitter tolerance	176C.6.4.6	Table 179-16	—

176C.6.4.1 Receiver signaling rate

The receiver shall comply with the requirements of 176C.6.4.5 and 176C.6.4.6 for any signaling rate in the range specified in Table 176C-4.

176C.6.4.2 Receiver difference ERL

The receiver difference ERL at TP5v is computed using the procedure in 163A.4.1.2 with the values in Table 176C-3 and Table 176C-9, with differential reference impedance equal to 92.5Ω , and with the value of T_{fx} equal to twice the delay from TP5 to TP5v. The reference value, $ERL^{(ref)}$, is calculated using the device package model of the receiver class to which the device adheres.

Receiver difference ERL at TP5v shall be greater than or equal to $dERL$ (min) specified in Table 176C-4.

176C.6.4.3 Receiver differential-mode to common-mode return loss

The differential-mode to common-mode return loss of the receiver shall meet Equation (176C-2). The differential-mode to common-mode return loss limit is illustrated by Figure 176C-5.

$$RL_{cd}(f) \geq \begin{cases} 25 - 20 \frac{f}{106.25} & 0.05 \leq f \leq 53.125 \\ 15 & 53.125 \leq f \leq 67 \end{cases} \quad (176C-2)$$

where

RL_{cd} is the differential-mode to common-mode return loss in dB
 f is the frequency in GHz

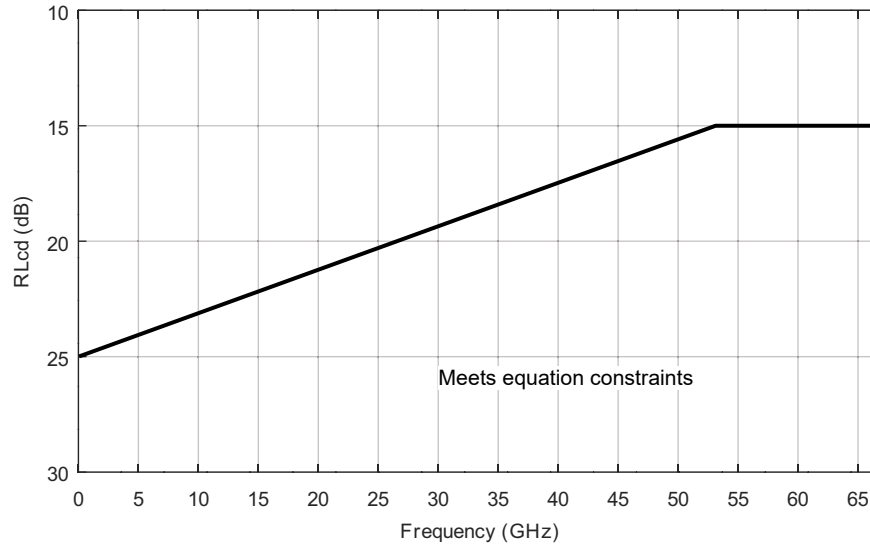


Figure 176C-5—Receiver differential-mode to common-mode return loss limit

176C.6.4.4 Error ratio requirements for receiver tests

A receiver on each lane is required to meet the expected BLER specified in 176C.2, using the test method in either 174A.9.5 or 174A.9.6 with parameters provided in Table 176C-5. The error mask $H_{\max}(k)$ to be used in the method of 174A.9.5 is provided in Table 176C-6.

NOTE—In order to predict whether a receiver meets the BLER requirement in a short test time, extrapolation of the measured histogram (see 174A.9.3) to $H_m(16)$ can be performed. One way of doing this is using a linear fit of the slope of $\log_{10}(H_m(k))$ for $k=0$ to n , where n is at least 4 and $H_m(k)=0$ for all $k \geq n$. For the linear fit, $H_m(n)$ is set to $3/N$ where N is the total number of test blocks, and the intercept $H_m(0)$ is set to 1. The fit is performed only for values of k that have $H_m(k)>0$.

Table 176C–5—Error ratio parameters

Parameter name	Value
p 200GAUI-1 400GAUI-2 800GAUI-4 1.6TAUI-8	1 2 4 8
BLER limit	1.45×10^{-11}
BER_{total}	2.921×10^{-4}
BER_{added}	2.841×10^{-5}
BER_{max}	8×10^{-6}

Table 176C–6—Receiver error mask

Test symbol errors per test block, k (see 174A.9.5)	Probability $H_{\text{max}}(k)$			
	$p = 1$	$p = 2$	$p = 4$	$p = 8$
1	4.2×10^{-2}	2.1×10^{-2}	1.1×10^{-2}	5.4×10^{-3}
2	9.1×10^{-4}	2.3×10^{-4}	5.8×10^{-5}	1.5×10^{-5}
3	1.3×10^{-5}	1.7×10^{-6}	2.1×10^{-7}	2.6×10^{-8}
4	1.4×10^{-7}	8.9×10^{-9}	5.5×10^{-10}	3.3×10^{-11}
5	1.2×10^{-9}	3.8×10^{-11}	1.2×10^{-12}	3.4×10^{-14}
6	8.8×10^{-12}	1.4×10^{-13}	2×10^{-15}	2.9×10^{-17}
7	5.4×10^{-14}	4.2×10^{-16}	3×10^{-18}	2×10^{-20}
8	2.9×10^{-16}	1.1×10^{-18}	3.9×10^{-21}	1.2×10^{-23}
9	1.4×10^{-18}	2.6×10^{-21}	4.4×10^{-24}	6.6×10^{-27}
10	5.9×10^{-21}	5.4×10^{-24}	4.5×10^{-27}	3.1×10^{-30}
11	2.3×10^{-23}	1.0×10^{-26}	4.1×10^{-30}	1.3×10^{-33}
12	8.2×10^{-26}	1.8×10^{-29}	3.4×10^{-33}	5×10^{-37}
13	2.7×10^{-28}	2.9×10^{-32}	2.6×10^{-36}	1.7×10^{-40}
14	8.1×10^{-31}	4.3×10^{-35}	1.9×10^{-39}	5.4×10^{-44}
15	2.3×10^{-33}	5.9×10^{-38}	1.2×10^{-42}	1.6×10^{-47}
16	6.1×10^{-36}	7.5×10^{-41}	7.3×10^{-46}	4.1×10^{-51}

176C.6.4.5 Receiver interference tolerance

Receiver interference tolerance is defined by the procedure described in 176C.6.4.5.1 through 176C.6.4.5.3.

Receiver interference tolerance is specified by two tests defined in Table 176C–7. Test L includes a low-loss channel. Test H includes a high-loss channel.

NOTE—The channel for Test H is dependent on the receiver class that a AUI component under test complies with.

A receiver shall meet the requirements of this test under both Test L and Test H conditions.

Table 176C–7—Receiver interference tolerance parameters

Parameter	Test L (low loss)		Test H (high loss)			Units
	Max	Target	Min	Max	Target	
BLER ^a	$< 1.45 \times 10^{-11}$					—
Insertion loss, IL_{dd} , at 53.125 GHz ^b		N/A ^c				
Receiver class A	12		25.5	26.5	—	dB
Receiver class B	12		22	23	—	dB
COM including effects of broadband noise ^d	—	3	—	—	3	dB

^a The BLER (see 174A.9) is measured with BER_{added} specified in 176C.2

^b IL_{dd} measured between TPt and TP5 (see Figure 93C–4) plus IL_{dd} of the specific package used by the test transmitter. See 178.9.3.4.2.

^c For test L, the connection between TP0v and TP5v (see Figure 93C–2) is a noise coupler without an additional ISI channel, and the insertion loss is not specified.

^d The COM value is the target for the receiver noise level calibration defined in 93C.2 step 7). The channel noise voltage applied in 93C.2 step 8) should be as close as practical to the value needed to produce the target COM. If higher amplitude values are used, this would demonstrate margin to the specification but this is not required for compliance.

176C.6.4.5.1 Test setup

The interference tolerance test is performed with the setup shown in Figure 93C–2 or its equivalent. Characterization and calibration of the various elements in the test setup uses the test configurations in Figure 93C–3, Figure 93C–4, Figure 93C–5, and Figure 93C–6. In each figure TP0v replaces TP0a and TP5v replaces TP5a. For Test L in Table 176C–7, the ISI channel in Figure 93C–2 and Figure 93C–4 is omitted.

The transmitter meets the requirements stated in 176C.6.3 with the added constraint that the difference steady-state voltage dV_f (see 178.9.2.5) is less than or equal to the minimum specified in Table 176C–2. The ISI channel emulates the frequency-dependent loss of the channel. The channel noise source emulates crosstalk, noise, and any other non-equalizable signal distortions that may be introduced by a transmitter or channel.

The ERL of the test setup in Figure 93C–4 measured at TP5 replica towards TPt meets the requirements in 176C.7.3.

The channel noise source has an adjustable output amplitude that is set according to the calibration procedure in 176C.6.4.5.2. The noise produced by the channel noise source is measured directly at the

output of the noise source (see Figure 93C–6). The noise is Gaussian with a crest factor of at least 5. The noise spectral density, $N_s(f)$, is normalized and constrained according to the relations in Equation (93C-1), where f_b is the signaling rate and f_l is 1 GHz. The average noise spectral density, $NSD_{average}$, is defined by Equation (93C-2).

A test system with a fourth-order Bessel-Thomson low-pass response with a 3 dB bandwidth of 60 GHz is to be used to measure the output of the transmitter and the channel noise source.

176C.6.4.5.2 Noise calibration

The output amplitude of channel noise source is set to yield a target test channel Channel Operating Margin (COM). The test channel COM is derived from the S-parameters for the test channel from TPt to TP5 replica $S^{(c)}$ measured using the test setup in Figure 93C–4. COM is calculated according to Annex 178A using the parameter values in Table 176C–9 and Table 176C–10 for the corresponding receiver class but with the following exceptions.

The transmitter model is determined in one of the following ways.

- If the transmitter is a device with known S-parameters and transition time T_r , these parameters are used instead of the transmitter device package model in 178A.1.4, and the $ILdd$ in Table 176C–7 is the $ILdd$ of the concatenated S-parameters of the device and the channel between TPt and TP5 replica. T_r should be provided as the value at the input of the device S-parameters network, as defined in 120G.3.1.4 but with no observation filter.
- If a calibrated instrument-grade transmitter is used, the TP0 to TP0v trace and the ISI channel in Figure 93C–2 and Figure 93C–3 are omitted, the TP0 to TP0v replica trace in Figure 93C–4 is omitted, and the transmitter model $S^{(t)}$ is omitted from Equation (178A–2). In this case, the $ILdd$ in Table 176C–7 is the $ILdd$ of the channel between TPt and TP5 replica. The filtered voltage transfer function $H^{(k)}(f)$ calculated in Equation (178A–13) uses the filter $H_i(f)$ defined in 178A.1.6.2, where T_r is the measured transmitter transition time (see 120G.3.1.4).

In addition, the following parameters are derived from measurements of the test transmitter collected using the test setup in Figure 93C–3.

- The parameter SNR_{TX} is set to the measured value of SNDR with $N_p = 55$.
- The parameter R_{LM} is set to the measured value of R_{LM} .
- The parameters A_{DD} and σ_{RJ} are calculated from the measured values of JH4u and JH_{RMS} using Equation (178–3) and Equation (178–4) respectively. If the transmitter jitter can be controlled, it is recommended to adjust jitter such that JH4u and JH_{RMS} are as close as practical to their limits in Table 176C–2.

The interference noise (see 178A.1.5) is derived from the S-parameters for the noise path from TPn to TP5 replica $S^{(n)}$ measured using the test setup shown in Figure 93C–5. For a given RMS noise amplitude σ_{ns} , the spectral density of the channel noise source $N_s(f)$ may be assumed to be $2\sigma_{ns}^2/f_b$ from DC to $f_b/2$ and 0 for frequencies greater than $f_b/2$. The spectral density measured at the output of the channel noise source may also be used as $N_s(f)$ for the calculation of COM and, in such cases, the value of σ_{ns}^2 is the integral of $N_s(f)$ over frequency.

COM is calculated using both Test 1 and Test 2 device package model transmission line lengths listed in Table 176C–9 on the receiver side. The value of COM is taken as the lower of the two calculated values. The value of σ_{ns} that yields the target value of COM specified in Table 176C–7 is determined (via iterative calculation or by other means). The amplitude of the channel noise source is then configured so that the measured RMS value at its output matches σ_{ns} .

176C.6.4.5.3 Test procedure

The receiver of each lane is tested using the following procedure.

- 1) Calibrate the amplitude of the channel noise source using the procedure defined in 176C.6.4.5.2.
- 2) The receiver under test configures the test transmitter equalizer coefficients and precoding to the settings it would select using the ILT function described in 179.8.9.
- 3) Configure the test transmitter to transmit either scrambled idle or PRBS31Q. During the test, the transmitters in all lanes of the AUI component under test transmit either scrambled idle or PRBS31Q, with equalization turned off (PRESET 1 condition).

A receiver shall meet the error ratio requirements defined in 176C.6.4.4 under the conditions specified for this test for the receiver class that the AUI component complies with, for both Test L and Test H in Table 176C–7.

176C.6.4.6 Receiver jitter tolerance

Receiver jitter tolerance is defined by the procedure described in 176C.6.4.6.1 and 176C.6.4.6.2.

176C.6.4.6.1 Test setup

The test setup shown in Figure 178–6, or its equivalent, is used. The test channel meets the insertion loss requirement for Test H in Table 176C–7.

176C.6.4.6.2 Test procedure

The jitter tolerance test procedure is similar to that of 176C.6.4.5, with the following exceptions:

- For each case in Table 179–16, the synthesizer frequency is set to the specified jitter frequency and the synthesizer output amplitude is adjusted until the specified peak-to-peak jitter amplitude for that frequency is measured at TP0v.
- The test channel calibration (see 176C.6.4.5.2) is performed with the transmitter output set to the jitter frequency and amplitude for Case G of Table 179–16. The additive noise level obtained is used for all other cases.

The receiver under test shall meet the error ratio requirements defined in 176C.6.4.4 for each pair of jitter frequency and peak-to-peak amplitude values listed in Table 179–16.

176C.6.4.7 Transmitter output amplitude tolerance

Transmitter output amplitude tolerance of a receiver is defined as the maximum transmitter steady-state voltage that the receiver can tolerate, using the Test L (low loss) test channel specified in Table 176C–7, such that it satisfies the error ratio requirements specified in 176C.6.4.4 when the PMD transmit function operates in DATA mode (see 179.8.2).

The transmitter steady-state voltage is measured as specified in 179.9.4.2.2 at the output of the pattern generator used in the test. The differential peak-to-peak output voltage of the pattern generator does not exceed the maximum value specified in Table 176C–2.

The pattern generator is initially configured to transmit training frames as defined in 178B.7.3 with transmit equalization at the INITIALIZE setting (see Table 179–8). During the initialization period, the device under test (DUT) may further configure the pattern generator to the transmit equalizer and precoding settings it would select using the ILT function (see 176C.3). The settings may be communicated via the ILT function

or by other means. After the pattern generator has been configured, the BLER is measured as specified in 174A.9 or 174A.11.

NOTE—Steady-state voltage is defined with PRESET 1. It is not initially generated by a transmitter, due to the INITIALIZE setting in Table 179–8. The receiver is not required to tolerate PRESET 1 unless it specifically requests it. The voltage observed at the receiver input is further attenuated by the test channel. Therefore, the receiver is not required to tolerate the maximum transmitter output voltage at its input.

176C.7 Channel characteristics

Table 176C–8 provides a summary of the channel characteristics for 200GAUI-1 C2C, 400GAUI-2 C2C, 800GAUI-4 C2C and 1.6TAUI-8 C2C interfaces, and references to the subclauses addressing each parameter. Unless otherwise indicated, the channel is defined between TP0 and TP5.

Table 176C–8—Channel characteristics summary

Description	Reference	Value	Unit
Minimum COM	176C.7.1	3	dB
Maximum insertion loss from TP0d to TP5d, IL_{dd} , at 53.125 GHz (recommended)	176C.7.2	32	dB
Minimum ERL ^a	176C.7.3.1	9.7	dB
Differential-mode to common-mode return loss, RL_{cd}	176C.7.3.3	Table 176C–12	dB
Minimum channel signal to common-mode ratio, $SCMR_{CH}$	176C.7.3.2	17	dB
Maximum AC-coupling 3 dB corner frequency	176C.7.5	250	kHz

^aChannels with a COM greater than 4 dB are not required to meet the minimum ERL.

176C.7.1 Channel Operating Margin

The Channel Operating Margin (COM) is computed using the procedure in 178A.1 and the parameters in Table 176C–9 and Table 176C–10. COM is calculated with the parameters for the transmitter and receiver classes that the channel under test is intended to support. COM is calculated twice, Test 1 and Test 2, with different values of device package model transmission line length z_p . COM shall be greater than or equal to 3 dB for both Test 1 and Test 2. This minimum value allocates margin for practical limitations on the receiver implementation as well as the allowed transmitter equalization coefficients.

NOTE—The parameters and values in Table 176C–9 and Table 176C–10 correspond to behavioral models of transmitters and receivers that are compliant to the C2C component specifications in this clause. The purpose of these parameters and values is to compute COM, a channel metric, and they do not represent requirements for transmitter and receiver implementations. It is expected that the C2C component specifications in this clause can be met with a variety of approaches to transmitter and receiver implementation.

Table 176C-9—Device termination and package parameters

Parameter	Symbol	Value	Units
Device termination model			
Single-ended device capacitance for stage 1	$C_d^{(1)}$	40×10^{-6}	nF
Single-ended device capacitance for stage 2	$C_d^{(2)}$	90×10^{-6}	nF
Single-ended device capacitance for stage 3	$C_d^{(3)}$	110×10^{-6}	nF
Single-ended device series inductance for stage 1	$L_s^{(1)}$	0.13	nH
Single-ended device series inductance for stage 2	$L_s^{(2)}$	0.15	nH
Single-ended device series inductance for stage 3	$L_s^{(3)}$	0.14	nH
Single-ended bump capacitance	C_b	30×10^{-6}	nF
Device package model, class A			
Transmission line parameter γ_0	γ_0	5×10^{-4}	1/mm
Transmission line parameter a_1	a_1	8.9×10^{-4}	ns ^{1/2} /mm
Transmission line parameter a_2	a_2	2×10^{-4}	ns/mm
Transmission line parameter τ	τ	6.141×10^{-3}	ns/mm
Transmission line 1 length, Test 1	$z_p^{(1)}$	33	mm
Transmission line 1 length, Test 2	$z_p^{(1)}$	12	mm
Transmission line 1 characteristic impedance	$Z_c^{(1)}$	87.5	Ω
Transmission line 2 length	$z_p^{(2)}$	1.8	mm
Transmission line 2 characteristic impedance	$Z_c^{(2)}$	92.5	Ω
Device package model, class B			
Transmission line parameter γ_0	γ_0	5×10^{-4}	1/mm
Transmission line parameter a_1	a_1	6.5×10^{-4}	ns ^{1/2} /mm
Transmission line parameter a_2	a_2	2.93×10^{-4}	ns/mm
Transmission line parameter τ	τ	6.141×10^{-3}	ns/mm
Transmission line 1 length, Test 1, Tx / Rx	$z_p^{(1)}$	45 / 44	mm
Transmission line 1 length, Test 2, Tx / Rx	$z_p^{(1)}$	30 / 29	mm
Transmission line 1 characteristic impedance	$Z_c^{(1)}$	87.5	Ω
Transmission line 2 length	$z_p^{(2)}$	2	mm
Transmission line 2 characteristic impedance	$Z_c^{(2)}$	95	Ω
Transmission line 3 length	$z_p^{(3)}$	1.3	mm
Transmission line 3 characteristic impedance	$Z_c^{(3)}$	100	Ω
Transmission line 4 length	$z_p^{(4)}$	1.5	mm
Transmission line 4 characteristic impedance	$Z_c^{(4)}$	78	Ω
Single-ended package capacitance at package-to-board interface	C_p	40×10^{-6}	nF
Single-ended reference resistance	R_0	46.25	Ω
Single-ended transmitter termination resistance	$R_d^{(t)}$	46.25	Ω
Single-ended receiver termination resistance	$R_d^{(r)}$	46.25	Ω
NOTE—The device package model parameters are based on the reference impedance R_0 .			

Table 176C–10—COM parameter values

Parameter	Symbol	Value	Units
Signaling rate	f_b	106.25	GBd
Transmitter differential peak output voltage			
Victim	A_v	0.385	V
Far-end aggressor	A_{fe}	0.385	V
Near-end aggressor	A_{ne}	0.481	V
Transmitter transition time	T_r	0.004	ns
Receiver 3 dB bandwidth	f_r	$0.55 \times f_b$	GHz
Transmitter equalizer, coefficient –3	$c(-3)$	0	—
Transmitter equalizer, coefficient –2	$c(-2)$		
Minimum value		0	
Maximum value		0.14	—
Step size		0.02	
Transmitter equalizer, coefficient –1	$c(-1)$		
Minimum value		–0.34	
Maximum value		0	—
Step size		0.02	
Transmitter equalizer, coefficient 0	$c(0)$		
Minimum value		0.54	—
Transmitter equalizer, coefficient 1	$c(1)$		
Minimum value		–0.2	
Maximum value		0	—
Step size		0.02	
Continuous time filter			
Zero 1 frequency for $g_1 = 0$	f_{z1}	$f_b / 2.5$	GHz
Zero 2 frequency for $g_2 = 0$	f_{z2}	$f_b / 80$	GHz
Pole 1 frequency	f_{p1}	$f_b / 2.5$	GHz
Pole 2 frequency	f_{p2}	f_b	GHz
Pole 3 frequency	f_{p3}	$f_b / 80$	GHz
Gain 1	g_1		
Minimum value		–20	dB
Maximum value		0	dB
Step size		1	dB
Gain 2	g_2		
Minimum value		–6	dB
Maximum value		0	dB
Step size		1	dB
Number of signal levels	L	4	—
One-sided noise spectral density	η_0	7.5×10^{-9}	V ² /GHz
Transmitter signal-to-noise ratio	SNR_{TX}	33.5	dB
Number of samples per unit interval	M	32	

Table 176C–10—COM parameter values (*continued*)

Parameter	Symbol	Value	Units
Random jitter, RMS	σ_{RJ}	0.01	UI
Dual-Dirac jitter, peak	A_{DD}	0.02	UI
Noise-equivalent quantization bits	N_{qb}	6	
Quantization clip probability	P_{qc}	10^{-7}	
Receiver discrete-time equalizer parameters			
Number of pre-cursor taps	d_w	5	—
Number of fixed-position taps	N_{fix}	14 ^a	—
Number of floating tap groups	N_{wg}	2	—
Number of taps per floating tap group	N_{wf}	4	—
Highest allowed tap index	N_{max}	56 ^b	—
Normalized upper limit on feed-forward coefficient $w(j)$ $1 \leq j \leq d_w$ and $d_w+2 \leq j \leq N_{fix}$ $N_{fix} < j$	$w_{max}(j)$	0.7 0.05	— —
Normalized lower limit on feed-forward coefficient $w(j)$ $1 \leq j \leq d_w$ and $d_w+2 \leq j \leq N_{fix}$ $N_{fix} < j$	$w_{min}(j)$	–0.7 –0.05	— —
Number of feedback taps	N_b	1	—
Normalized upper limit on feedback coefficient $b(j)$	$b_{max}(j)$	0.85	—
Normalized lower limit on feedback coefficient $b(j)$	$b_{min}(j)$	0	—
Level separation mismatch ratio	R_{LM}	0.95	—
Target detector error ratio	DER_0	0.67×10^{-5}	—
Include maximum likelihood sequence detection		no	—

^a The number of fixed-position post-cursor taps is 8.

^b The latest post-cursor position for a floating tap is 50.

176C.7.2 Channel insertion loss (recommended)

The recommended maximum insertion loss, IL_{dd} , for the channel between TP0d and TP5d is specified in Table 176C–8.

176C.7.3 Channel return loss specifications

176C.7.3.1 Channel ERL

The channel ERL at TP0 and at TP5 is computed using the procedure in 93A.5 with the values in Table 176C–11 and with differential reference impedance equal to 92.5 Ω . Parameters that do not appear in

Table 176C–11 take values from Table 176C–9. Channel ERL at TP0 and at TP5 shall meet the requirement in Table 176C–8.

Table 176C–11—Channel ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.005	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	4000	UI
Equalizer length associated with reflection signal	N_{bx}	16	UI
Time-gated propagation delay	T_{fx}	0	ns
Tukey window flag	tw	1	—

176C.7.3.2 Channel modal ERL

Channel modal ERL (ERL_{CD}) is the ERL computed according to 93A.5 with the following exceptions:

- For channel input ERL_{CD} , $S_{ii}(f)$ (see 93A.5.1) is replaced with the channel $S_{CD11}(f)$.
- For channel output ERL_{CD} , $S_{ii}(f)$ is replaced with the channel $S_{CD22}(f)$.

The values of $S_{CD11}(f)$ and $S_{CD22}(f)$ are defined in 178A.1.3. Modal ERL is computed using the values in Table 176C–10 and Table 176C–11. The differential reference impedance is 92.5 Ω and the common-mode reference impedance is 23.125 Ω .

176C.7.3.3 Channel modal return loss

The channel modal return loss specification is differential-mode to common-mode.

The TP0 to TP5 channel differential-mode to common-mode return loss is satisfied by either meeting the requirement of Equation (178–6) or by having ERL_{CD} higher than the value in Table 176C–12.

Table 176C–12—Channel modal return loss specifications

RL mask equation	RL mask figure	ERL_{CD} minimum (dB)
Equation (178–6)	Figure 178–8	17

176C.7.4 Channel signal to common-mode ratio

Channel signal to common-mode ratio, $SCMR_{CH}$, is calculated using the procedure in 178A.2 with $P_{peak} = 10^{-7}$ and other parameter values taken from Table 176C–10. $SCMR_{CH}$ shall be greater than or equal to 17 dB.

176C.7.5 AC Coupling

The channel AC-coupling specifications are identical to those of 178.10.4.

176C.8 Management variables

AUI component control and status variables intended to be accessible via a management system are listed in Table 179–26 and Table 179–27. Additional variables associated with the APSU functions are listed in Table 178B–3 and Table 178B–4.

NOTE—Although the variable names and descriptions in Table 179–26 and Table 179–27 include the term “PMD”, the same variables names are used in AUI components.

If the MDIO Interface (see Clause 45) is implemented, AUI component control and status variables shall be mapped to MDIO registers as shown in Table 179–26, Table 179–27, Table 178B–3, and Table 178B–4, with the following exceptions:

- For a host-side AUI component (see Figure 176C–3), which is the lower AUI component (see 178B.3), the MMD number is different.
- For a network-side AUI component (see Figure 176C–3), which is the upper AUI component (see 178B.3), all register numbers are offset by 4000 (see 45.2.1.273) and if not in the same package as the PMD the MMD number is different.

If the MDIO Interface is not implemented, an alternate mechanism to access management variables shall be provided.

176C.9 Protocol implementation conformance statement (PICS) proforma for Annex 176C, Chip-to-chip Attachment Unit Interfaces 200GAUI-1 C2C, 400GAUI-2 C2C, 800GAUI-4 C2C, and 1.6TAUI-8 C2C⁶

176C.9.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Annex 176C, Chip-to-chip Attachment Unit Interfaces 200GAUI-1 C2C, 400GAUI-2 C2C, 800GAUI-4 C2C, and 1.6TAUI-8 C2C, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

176C.9.2 Identification

176C.9.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

176C.9.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3dj-202x, Annex 176C, Chip-to-chip Attachment Unit Interfaces 200GAUI-1 C2C, 400GAUI-2 C2C, 800GAUI-4 C2C, and 1.6TAUI-8 C2C
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3dj-202x.)	

Date of Statement	
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⁶*Copyright release for PICS proformas:* Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

176C.9.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
NOL	Number of differential AC-coupled lanes	176C.1	One independent data path in each direction for 200GAUI-1 C2C, two independent data paths in each direction for 400GAUI-2 C2C, four independent data paths in each direction for 800GAUI-4 C2C, and eight independent data paths in each direction for 1.6TAUI-8 C2Ce	M	Yes []
*CHAN	Channel	176C.7	Items marked with CHAN include channel specifications not applicable to a PHY manufacturer	O	Yes [] No []
*PCSPKG	PMA in same package as PCS.			O	Yes [] No []
*MD	MDIO capability	176C.8	Clause 45 MDIO is implemented with the management variable mapping in 178.13	!CHAN: O	Yes [] No []
AUIPSU	APSU functions in AUI-C2C	176C.6.3	PSU functions are implement in the AUI-C2C.	AUI200 G:M	Yes [] N/A []

176C.9.4 PICS proforma tables for Chip-to-chip Attachment Unit Interfaces 200GAUI-1 C2C, 400GAUI-2 C2C, 800GAUI-4 C2C, and 1.6TAUI-8 C2C

176C.9.4.1 Transmitter

Item	Feature	Subclause	Value/Comment	Status	Support
TC1	Signaling rate, each lane, PMA is in same package as PCS	176C.6.3	Per Table 176C–2	PCSP KG:M	Yes [] N/A []
TC2	Signaling rate, each lane, PMA is not in same package as PCS	176C.6.3	Derived from PMA input lanes	!PCSP KG:M	Yes [] N/A []
TC3	Peak-to-peak differential output voltage, transmitter enabled	176C.6.3	Per Table 176C–2	M	Yes []
TC4	Peak-to-peak differential output voltage, transmitter disabled	176C.6.3	Per Table 176C–2	M	Yes []
TC5	DC common-mode voltage	176C.6.3	Per Table 176C–2	M	Yes []
TC6	Low-frequency peak-to-peak AC common-mode voltage	176C.6.3	Per Table 176C–2	M	Yes []
TC7	Signal to AC common-mode noise ratio	176C.6.3	Per Table 176C–2	M	Yes []
TC8	Difference ERL	176C.6.3	Per Table 176C–2	M	Yes []
TC9	Common-mode to common-mode return loss	176C.6.3	Per Table 176C–2	M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
TC10	Output waveform coefficient limits	176C.6.3	Per Table 176C–2	M	Yes []
TC11	Output waveform coefficient step size	176C.6.3	Per Table 176C–2	M	Yes []
TC12	Output jitter	176C.6.3	Per Table 176C–2	M	Yes []
TC13	Difference signal-to-noise-and-distortion ratio	176C.6.3	Per Table 176C–2	M	Yes []
TC14	Transmitter precoder request	176C.3		O	Yes [] No []
TC15	Difference steady-state voltage	176C.6.3	Per Table 176C–2	M	Yes []
TC16	Difference linear fit pulse peak ratio	176C.6.3	Per Table 176C–2	M	Yes []
TC17	Level separation mismatch ratio	176C.6.3	Per Table 176C–2	M	Yes []

176C.9.4.2 Receiver

Item	Feature	Subclause	Value/Comment	Status	Support
RC1	Difference ERL	176C.6.4	Per Table 176C–4	M	Yes []
RC2	Differential-mode to common-mode return loss	176C.6.4	Per Table 176C–4	M	Yes []
RC3	Interference tolerance	176C.6.4.5	Per Table 176C–7	M	Yes []
RC4	Jitter tolerance	176C.6.4.6	Per Table 179–15	M	Yes []
RC5	Transmitter equalization feedback	176C.6.3.1		O	Yes [] No []

176C.9.4.3 Channel

Item	Feature	Subclause	Value/Comment	Status	Support
CC1	Channel operating margin (COM)	176C.7.1	Per Table 176C–8	CHAN:M	Yes [] N/A []
CC2	Channel insertion loss	176C.7.2	Per Table 176C–8	CHAN:O	Yes [] No [] N/A []
CC2	Channel ERL	176C.7.3	Per Table 176C–8	CHAN:M	Yes [] N/A []
CC4	Channel AC-coupling	176C.7	Per Table 176C–8	CHAN:M	Yes [] N/A []

176C.9.4.4 Management

Item	Feature	Subclause	Value/Comment	Status	Support
M1	Alternate access to management variables is provided	176C.8		!MD:M	Yes [] N/A []
M2	Mapping of MDIO control bits and MDIO status bits	176C.8		MD:M	Yes [] N/A []